

BSCS FINAL PROJECT PHASE 1

Smart Eye Testing and Wellness Guidance



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Software Requirements Specification

Phase 1 Smart Eye Testing and Wellness Guidance

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Revision History

Name	Date	Reason For Changes	Version
Nayab Irfan	17/11/25	Updated project title from “ Smart Eye Testing & Wellness Guidance ” to “ Visuar ” for clarity and branding	v0.0

Previous Phases Feedback

Dear Students,

Following is the feedback regarding your Idea Defence:

Sr. #	CS-122
Project Title	SMART EYE TESTING AND WELLNESS GUIDANCE

Similar products already exist. Well explained.

Prepare your Proposal Document in the light of comments provided. Last date to submit the Proposal Document is **Tuesday, September 16, 2025** not later than **04:00 PM**. The proposal document should be prepared using the template available at http://smrms.ucp.edu.pk/All_Phases_Templates.html. Submit the hard copy to Project Office in F-304 (Main Building).

When submitting the hard copy of the proposal document, use the duly signed Group Allocation Yellow Page (available at <http://smrms.ucp.edu.pk/Forms.html>) as the first page of the submission document. Also get the proposal tape bind before submission.

Abstract

Visuar is an AI-driven vision testing and wellness guidance system designed to make eye-health screening accessible, low-cost, and available to anyone through a standard webcam. The project addresses a critical gap in regions where professional eye care is limited, expensive, or inconvenient—leading to late detection of common refractive issues such as myopia, hyperopia, astigmatism, presbyopia, and emerging digital eye strain caused by prolonged device use. More serious conditions, including early indicators of glaucoma, diabetic retinopathy, hypertensive retinopathy, thyroid-related eye changes, and neurological abnormalities, also often go unnoticed until symptoms worsen. Visuar provides an automated, interactive screening experience by combining computer vision, eye-tracking, and lightweight machine-learning pipelines. Using webcam-based landmark extraction, blink-rate measurement, gaze-stability tracking, color-differentiation tests, and focus-response analysis, the system evaluates a user's visual behavior and detects patterns associated with early vision problems and lifestyle-driven strain. The underlying solution draws upon multiple knowledge areas, including deep learning, computer vision, HCI, and software engineering, while integrating insights from publicly available datasets such as RT-GENE, RT-BENE, NUIG EyeGaze, and custom gaze-direction datasets. The expected outcome is a functional prototype capable of performing at-home non-clinical eye assessments, generating personalized wellness reports, and highlighting potential risk indicators that may require professional consultation. Visuar aims to democratize eye-health screening by providing an intelligent, accessible, and research-backed platform that supports preventive care and promotes early awareness of visual and systemic health concerns.

1. Introduction

1.1 Review of Related Literature

Recent research demonstrates that vision assessment and ocular health prediction can now be performed using commodity devices such as webcams, smartphones, and low-cost sensors. Digital vision testing platforms such as Peek Vision and EyeQue introduced smartphone-based eye exams but lack personalized recommendations, multilingual support, and intelligent behavior analysis. Recent research on AI in ophthalmology focuses on retinal image analysis using CNNs, but these solutions require high-quality fundus images and expensive hardware, making them unsuitable for mass testing. Smartphone-based screening tools such as Vis-Screen (MDPI) validate the feasibility of conducting refractive error tests and visual acuity assessments remotely [1]. Studies such as the Prevalence of Computer Vision Syndrome (Nature Scientific Reports) highlight the growing need for early detection of digital eye strain, especially among young populations with heavy screen exposure [2]. Machine-learning-driven systems like RT-GENE and RT-BENE provide robust blink, gaze, and eye-feature datasets enabling real-time estimation of ocular behavior [8,11]. Additional research, including Eye Fatigue Detection via Eye-Tracking Signals (MDPI Electronics), confirms the role of blink rate, saccades, gaze stability, and fixation patterns in determining eye strain and attention deficits [4]. Newer AI works (ESPIM — Eye-Strain Probation Model) demonstrate that eye movement data can predict ergonomic fatigue and interaction patterns [3]. Overall, literature strongly supports that webcam-based vision testing, gaze analytics, and AI-enhanced diagnostics can reliably detect early markers of refractive errors, eye strain, and systemic health risks [1,2,3,4,8].

1.2 Problem Statement

In regions such as Pakistan, timely access to basic eye care is limited due to high consultation fees, scarcity of qualified optometrists, and lack of awareness. Common conditions such as myopia, hyperopia, astigmatism, presbyopia, digital eye strain, and even early indicators of diabetic retinopathy, glaucoma, and hypertensive retinopathy often remain undetected until symptoms become severe. Traditional screening methods; Snellen charts, Ishihara plates, and clinic-based refraction tests; require in-person visits, controlled lighting, and trained personnel, making early screening inaccessible for:

- children,
- elderly individuals,
- busy working professionals,
- rural populations.

This creates a critical gap: there is no intelligent, low-cost, at-home screening solution capable of analyzing webcam footage to assess vision quality and early health risks.

1.3 Proposed Solution

Visuar is a smart, AI-driven eye testing and wellness guidance system designed to make vision assessment accessible from any device equipped with a webcam.

The system performs:

- Eye landmark extraction (MediaPipe / custom CNN)
- Blink rate and blink quality analysis
- Gaze direction tracking & stability checks
- Color differentiation tests
- Focus and accommodation response assessment
- Early indicators of refractive errors
- Screening for digital eye strain
- Mapping visual patterns to systemic risks (diabetes, hypertension, thyroid, neurological issues)

After processing, Visuar generates a personalized eye wellness report, risk flags, and AI-powered recommendations. It can also produce a shareable summary for professional consultation.

1.4 Problem Scope

Visuar focuses on delivering non-clinical, initial-level screening, not medical diagnosis.

Included in Scope:

- Webcam-based vision behavior analysis
- Eye strain, focus stability, blink patterns
- Gaze stability & saccade smoothness
- Color perception & basic contrast tests
- AI-assisted refractive error indicators (approximate)
- Personalized recommendations
- User profile management and test history

Out of Scope:

- Full clinical refraction
- Prescription-grade eyeglass power
- Detection of retinal diseases using fundus images
- Medical-grade diagnosis

The system is intended as a first-line digital screening tool, not a replacement for an eye specialist.

1.5 Challenges

- Accurate eye tracking in varying lighting conditions
- Variability in webcam quality
- Noise and blur during movement
- Extracting reliable blink metrics from low-resolution videos
- Ensuring fairness for children, elderly, and users with glasses
- Designing multi-mode test flows (general, child, elderly)
- Handling privacy and data security for recorded data

- Integrating multiple analytical pipelines (gaze, blink, color tests, cognitive stress indicators)
- Creating an interpretable AI model for wellness recommendations

1.6 Knowledge Areas Required

- Database systems
- Machine Learning & Deep Learning
- Human-Computer Interaction (HCI)
- Web / Desktop App Development
- Software Engineering & Design Patterns
- Data Preprocessing & Feature Engineering
- Signal Processing for eye behavior
- UX/UI Design for health tech
- Python-based AI pipelines
- Model evaluation and dataset management
- Computer Vision (OpenCV, MediaPipe)
- NLP & LLMs (OpenAI, LangChain)
- Speech technologies (Whisper, Google TTS, ElevenLabs)
- Software engineering, APIs, microservices

1.7 Completeness Criteria

Component	Description	Weight
Webcam Vision Module	Real-time face, landmark, gaze & blink capture	20%
Test Modes & UI	General, Child, Elderly, Strain tests	15%
ML/AI Analysis Engine	Accuracy of gaze/blink/strain/refraction estimation	30%
Color + Focus Tests	Working Ishihara-like & contrast modules	10%
Report Generator	AI-driven recommendations & risk flags	15%
Data Storage / User Profile	Logging, tracking history	10%

Table 1: Completeness Criteria

- Working prototype of 3+ vision tests
- Voice-based interaction flow
- Partial eye-tracking implementation
- Report generation functioning
- Basic data storage working

1.8 Research Outcomes Nature of End Product

The final product will be a hybrid AI-driven assessment tool that:

- *Performs non-invasive, webcam-based eye screening*
- *Generates insights aligned with research-backed biomarkers*
- *Helps users understand strain levels, visual issues, and risks*
- *Serves as a foundation for future clinical-grade models*
- *Contributes a structured dataset of gaze/blink/focus patterns collected voluntarily*
- *Provides a practical, deployable software prototype*

Type of End Product:

A fully functional AI-powered screening application + research-backed documentation + dataset pipeline.

1.9 Learning Outcomes

Students will learn:

- *Learn dataset preprocessing and ML training techniques*
- *Clinical & Biomedical Understanding*
- *Medical Data Interpretation*
- *Healthcare Technology Development*
- *Machine-Learning & Clinical AI Skills*
- *Research & Scientific Communication*
- *Gain expertise in MediaPipe, OpenCV, and deep learning models*
- *Understand eye-tracking and gaze estimation methodologies*
- *Master front-end + back-end integration for real-time systems*
- *Interpret clinical and human-factors parameters such as blink metrics and focus behavior*
- *Understand SRS writing, UML modeling, and research documentation*
- *Improve teamwork, version control, and project planning skills*

1.10 Document Conventions

- *Italic text is used for emphasis or terminology.*
- *Font 11 Arial for body and 14 for subheading and 18 for main titles.*
- **Text** represents system modules, features, or component names.
- *UML diagrams follow standard UML notation.*
- *PlantUML syntax is used for modeling workflow and process diagrams.*
- *Technical terms related to ML (CNN, LSTM, landmarks, saccades, etc.) follow standard CV/AI literature conventions.*
- *All datasets referenced follow their original authors' naming conventions.*

2. Background Study and Literature Survey

Vision problems are among the most underdiagnosed health issues, especially in regions where access to regular eye checkups is limited. Many individuals remain unaware of early symptoms of myopia, eye strain, color vision deficiency, or fatigue until the condition interferes with daily activities. Traditional testing methods like Snellen charts and clinical vision screening require physical presence, controlled environments, and trained professionals, making them inaccessible for children, elderly individuals, and people in remote areas.

With advancements in computer vision, deep learning, and webcam-based eye-tracking, it is now possible to conduct meaningful visual assessments digitally. Recent studies (RT-GENE, NUIG EyeGaze, Harvard FairVision) show that machine learning can detect blink patterns, gaze direction, and visual anomalies with high accuracy [6,8,11]. Visuar builds upon this growing research space, integrating AI-driven analysis with accessible webcam tests to provide a low-cost and intelligent eye wellness solution.

3. Overall Description

3.1 Proposed Solution

*Visuar is an AI-powered, webcam-based vision screening and wellness guidance system designed to bring early-stage eye-health assessment into homes, schools, workplaces, and primary care settings. The system performs non-clinical, preliminary evaluations using real-time video analysis to measure blink behavior, gaze stability, focus accuracy, color perception, ocular fatigue levels, and visual attention patterns. The solution integrates ML-based feature extraction, temporal behavioral modeling, and health-risk interpretation to detect signs associated with refractive errors (**myopia, hyperopia, astigmatism, presbyopia**), digital eye strain, and wellness-related indicators such as fatigue, attention deficiency, and improper screen habits. While not a medical diagnostic device, Visuar provides data-driven insights, personalized recommendations, and early alerts encouraging users to seek professional care when concerning patterns are detected.*

VISUAR consists of three phases:

3.1.1 Interaction Phase

- *Whisper STT + Google/ElevenLabs TTS*
- *Collect lifestyle info*
- *Estimate approximate screen distance*

3.1.2 Vision Testing Phase

- *Contrast sensitivity*
- *Dynamic visual acuity*
- *Color differentiation*
- *Depth perception*
- *Reaction time*

- Eye tracking

3.1.3 Diagnosis & Recommendation Phase

- LLM interprets results
- Generates lens suggestions
- Recommendations
- PDF sent via Email

Block diagram (will be drawn in Phase 2).

3.2 User Classes and Characteristics

User Class	Characteristics	Special Considerations
General Users	Adults using the system for routine wellness checks	Varying levels of digital literacy; need simple UI/UX
Students/Children	Younger users prone to screen overuse, digital eye strain	Child-friendly instructions, larger fonts, game-like test flow
Elderly Users	May have age-related vision issues (presbyopia, cataracts, AMD)	High-contrast UI, slower instructions, larger icons
Office Workers	Extended screen exposure, high risk of CVS (Computer Vision Syndrome)	Frequent-use optimisation, ergonomic feedback
Healthcare Support Users	Optional use by optometrists or technicians for screening camps or triage	Ability to export reports and evaluate risk flags
Parents / Caregivers	Monitoring screen use and eye health of children	Shared accounts and periodical report summaries

Table 2: User Classes & Characteristics

- **General Users:** low technical knowledge, use voice.
- **Pharmacy/Clinic Staff:** perform repeated tests.
- **School Screening Administrators:** bulk testing.

3.3 Operating Environment

Visuar is designed to operate in commonly available consumer environments with minimal hardware requirements.

Hardware Environment

- *Any laptop or desktop computer with a built-in or external webcam*
- *Optional smartphone browser support (for future phases)*
- *Minimum 720p camera recommended for accurate landmark detection*

Software Environment

- *Web-browser based interface (primary)*
- *Supported browsers: Chrome, Edge, Firefox, Safari*
- *Backend running Python-based ML models (Flask/FastAPI)*
- *AI inference supported through ONNX, TensorFlow Lite, or server-side processing*

Network Requirements

- *Stable internet connection for real-time inference (cloud mode)*
- *Offline running possible for basic models (if supported hardware is used)*

Environmental Conditions

- *Adequate lighting for accurate eye detection*
- *User positioned 40–70 cm from the webcam*
- *Minimal head movement*

3.4 Design and Implementation Constraints

Webcam Quality Variability

- *Users may have low-resolution cameras, affecting landmark accuracy.*
- *Models must be robust to poor lighting and noise.*

Real-Time Processing Limitations

- *Eye-tracking, blink detection, and gaze stability must run at acceptable FPS on consumer hardware.*

Non-Diagnostic Nature

- *System must clearly indicate that it is a wellness tool, not a medical diagnostic instrument.*

Data Privacy and Security

- *No raw facial video may be stored without explicit consent.*
- *GDPR and HIPAA-inspired guidelines must be considered depending on deployment.*

Dataset Restrictions

- *Clinical datasets (e.g., DR, AMD, Glaucoma) had licensing and ethical usage limitations.*
- *Model must generalize across demographic variations.*

Browser Permissions

- *Continuous webcam access must follow browser security rules.*

User Behavior Variability

- *Children or elderly may move excessively or fail to follow instructions consistently, hence voice and mic are required.*

3.5 Assumptions and Dependencies

Assumptions

- *The user has a functioning webcam and stable lighting conditions.*
- *Users will follow verbal/visual instructions during assessment.*
- *The environment is quiet and stationary enough for accurate screen-based testing.*
- *The system is used for wellness screening only, not clinical diagnosis.*
- *Future training data will be available to improve accuracy (e.g., blink datasets, gaze datasets, disease datasets).*

Dependencies

- *Availability of ML models such as:*
 - *MediaPipe for landmark detection*
 - *RT-BENE/RT-GENE datasets for blink & gaze*
 - *NUIG EyeGaze dataset for calibration robustness*
 - *Harvard FairVision disease datasets (for future advisory models)*
- *Availability of cloud compute or local GPU for inference.*
- *Functioning internet connectivity if cloud inference mode is used.*
- *Integration with browser APIs for video capture.*
- *Correct installation of Python packages and ML dependencies.*

4. Functional Requirements

4.1 At-Home Vision Screening

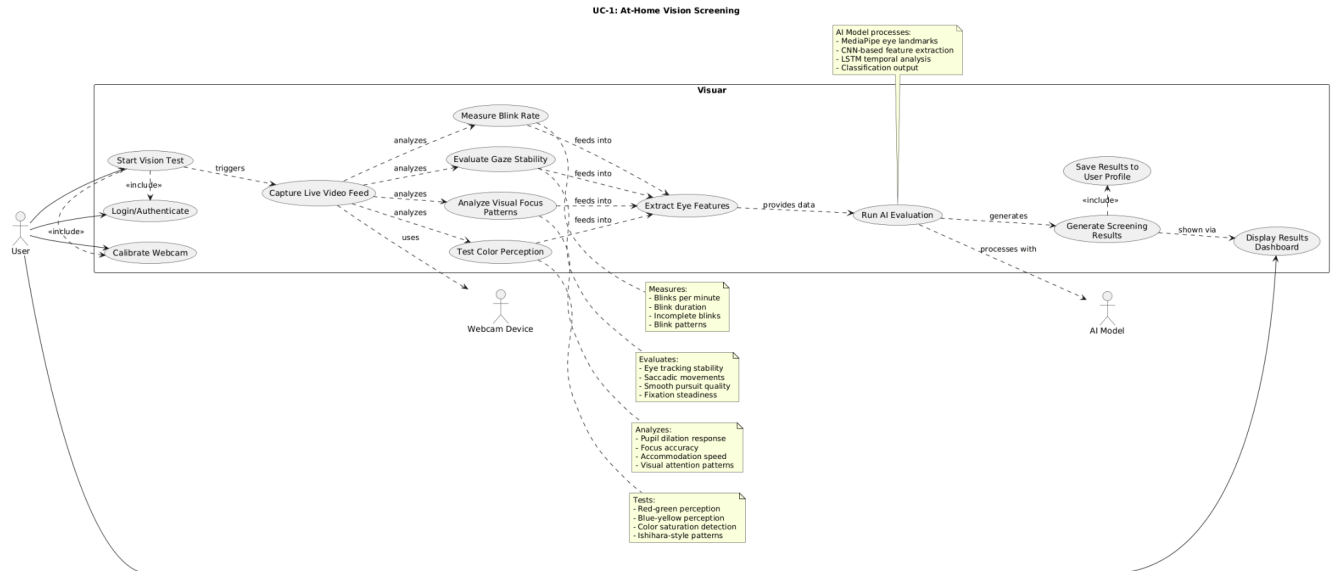


Figure 1: At-Home Vision Screening

Identifier	UC-1	
Purpose	Allow the user to perform a guided webcam-based vision test	
Priority	High	
Pre-conditions	Webcam active; stable lighting; user logged in	
Post-conditions	Vision test completed; results generated	
Typical Course of Action		
S#	Actor Action	System Response
1	User starts "Vision Test"	System activates webcam; checks lighting and face visibility
2	User follows on-screen instructions	System displays guided test steps (gaze, blink, color tests)
3	User performs gaze movements and reacts to stimuli	System records pupil movement, blink rate, tracking behavior
4	User completes the final step	System analyzes captured data using AI model

5	—	System generates Vision Screening Summary
Alternate Course of Action		
S#	Actor Action	System Response
1	User moves out of frame	System prompts the user to reposition their face
2	Lighting is poor	System requests the user to improve lighting
3	User cancels test	System discards incomplete data and returns home

Table 3: At-Home Vision Screening

4.2 Early Detection of Eye Strain

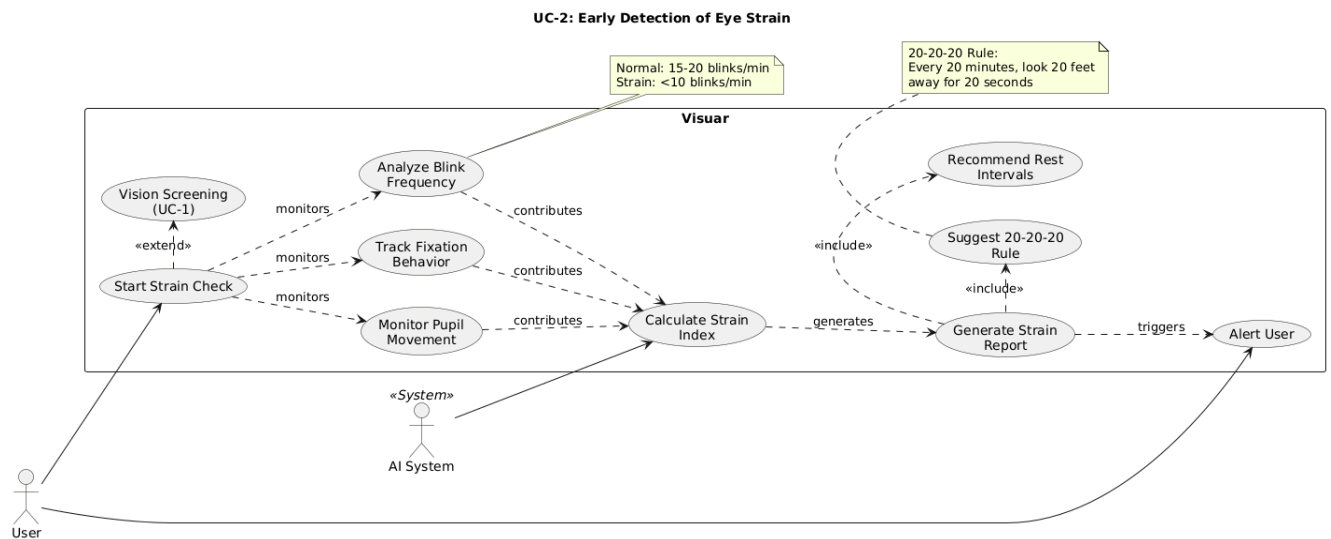


Figure 2: Early Detection of Eye Strain

Identifier	UC-2	
Purpose	Detect early signs of digital eye strain	
Priority	High	
Pre-conditions	Vision test active; webcam functioning	
Post-conditions	Eye strain status provided to user	
Typical Course of Action		
S#	Actor Action	System Response

1	User begins "Eye Strain Check"	System initiates short behavioral scan
2	User focuses on screen	System collects blink frequency and gaze stability
3	User follows visual cues	System detects micro-fixations and fatigue signals
4	User completes session	System analyzes downloaded data
5	—	System displays Eye Strain Risk Rating and guidance
Alternate Course of Action		
S#	Actor Action	System Response
1	Excessive blinking (allergies/dry eyes)	System alerts: "Abnormal patterns detected — retest recommended."
2	User stops mid-test	System pauses test and offers a resume option

Table 4: Early Detection of Eye Strain

4.3 Myopia | Hyperopia Indication

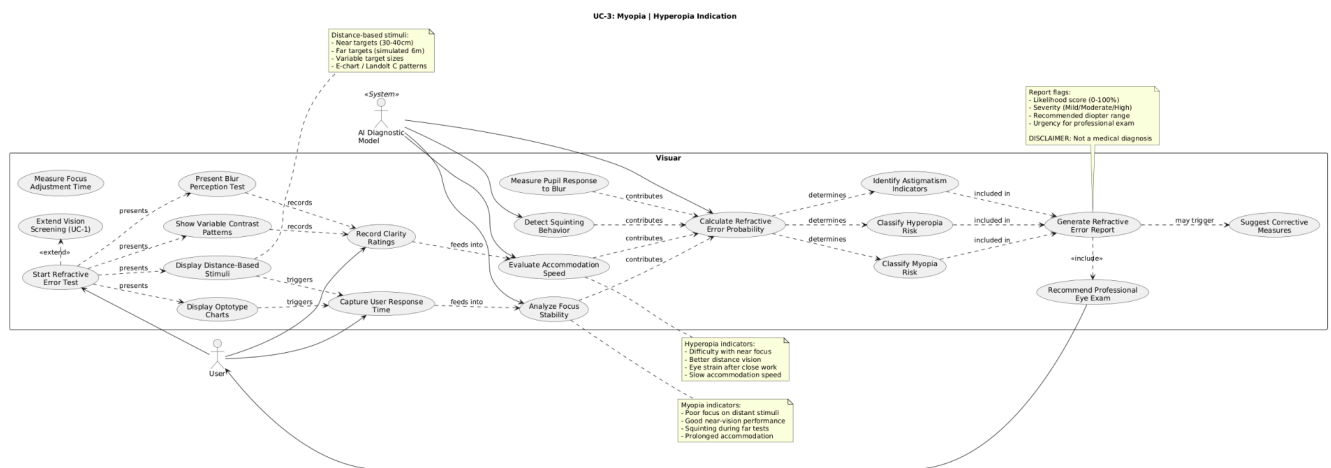


Figure 3: Myopia / Hyperopia Indicator Test

Identifier	UC-3
Purpose	Identify possible refractive error indicators
Priority	High
Pre-conditions	Webcam active; user positioned properly

Post-conditions		Refractive error indicators flagged
Typical Course of Action		
S#	Actor Action	System Response
1	User begins " Distance Focus Test "	System displays near and far visual stimuli
2	User focuses on changing targets	System measures focus lag and tracking performance
3	User reports blur (optional)	System records subjective responses
4	User completes test	AI analyzes signals for myopia/hyperopia indicators
5	—	System shows Possible Refractive Error Indicator
Alternate Course of Action		
S#	Actor Action	System Response
1	User too close/far	System prompts correct viewing distance
2	Webcam resolution too low	System requests switching devices

Table 5: Myopia – Hyperopia Indicator Test

4.4 Elderly Vision Check

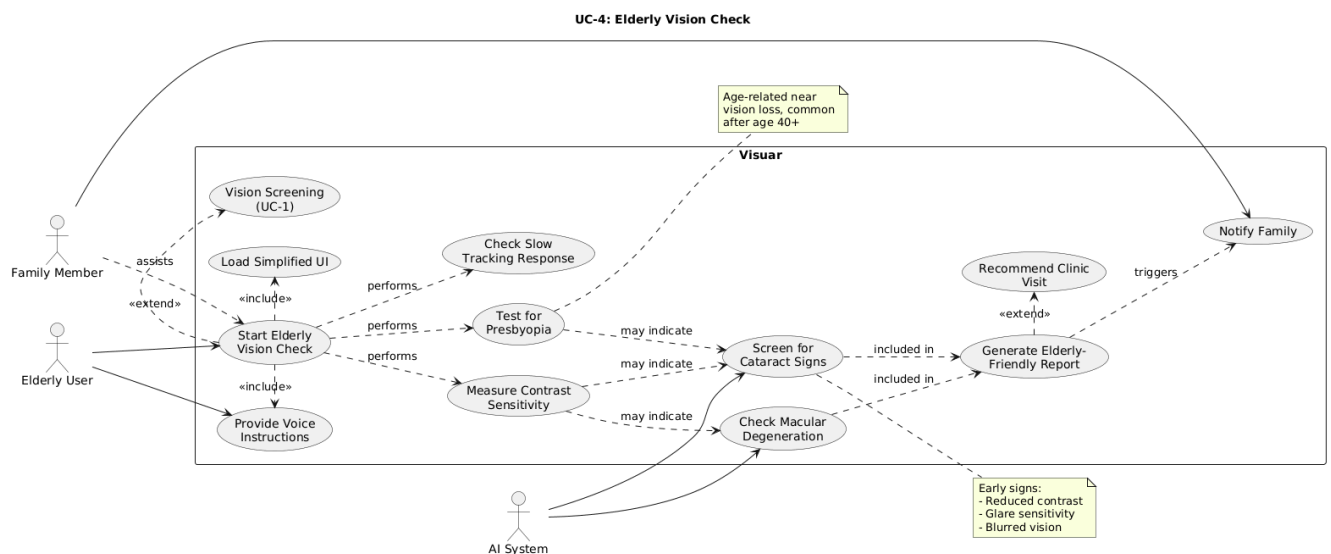


Figure 4: Elderly Vision Check

Identifier	UC-4	
Purpose	Provide simplified vision testing for elderly users	
Priority	Medium	
Pre-conditions	User selects "Elder Mode" ; webcam active	
Post-conditions	Simplified vision results generated	
Typical Course of Action		
S#	Actor Action	System Response
1	User selects "Elderly Vision Check"	System enables large-text, slow-paced interface
2	User follows easy visual cues	System detects presbyopia indicators and tracking latency
3	User confirms prompts	System records inputs
4	User completes test	System generates simplified results
Alternate Course of Action		
S#	Actor Action	System Response
1	User struggles with instructions	System switches to voice-only guidance
2	User pauses or becomes confused	System offers restart/resume options

Table 6: Elderly Vision Check

4.5 Child Vision Monitoring

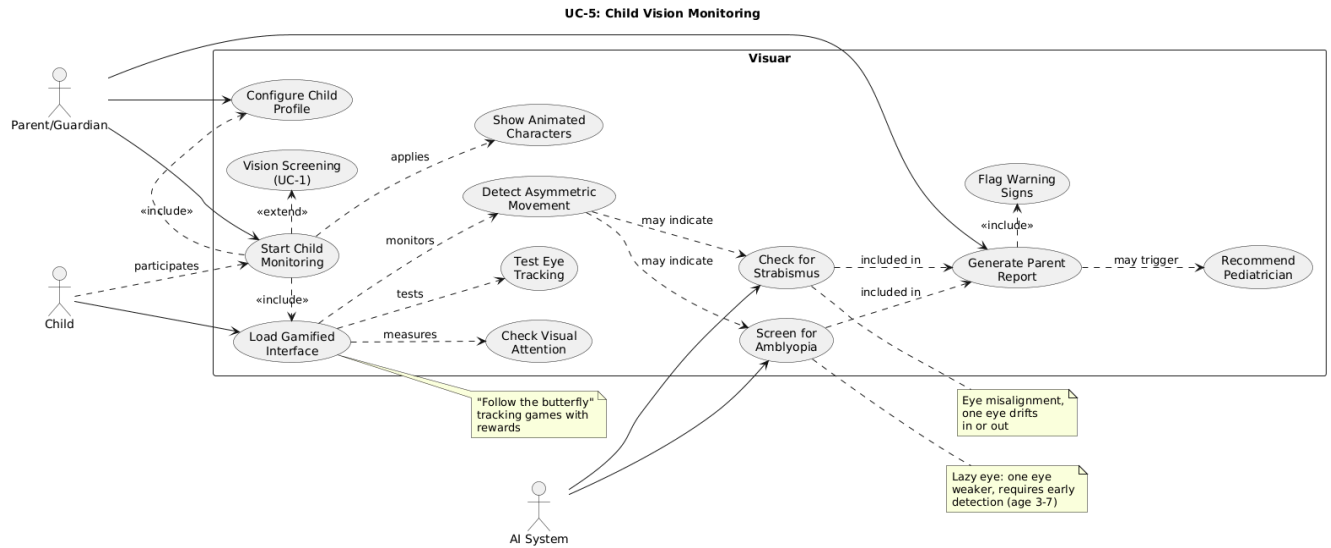


Figure 5: Child Vision Monitoring

Identifier		UC-5
Purpose		Help parents detect early childhood vision development issues
Priority		Medium
Pre-conditions		Child visible to camera; parent supervising
Post-conditions		Child Vision Summary generated
Typical Course of Action		
S#	Actor Action	System Response
1	Parent selects "Child Vision Check"	System activates child-friendly animated interface
2	Child follows animated objects	System analyzes tracking symmetry and response time
3	Child completes test	System evaluates developmental visual patterns
4	—	System generates Child Vision Health Insight
Alternate Course of Action		
S#	Actor Action	System Response
1	Child moves excessively	System prompts parent to stabilize the child

2	Child stops watching	System pauses and restarts animations
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Table 7: Child Vision Monitoring

4.6 Eye Health Insights for Professionals

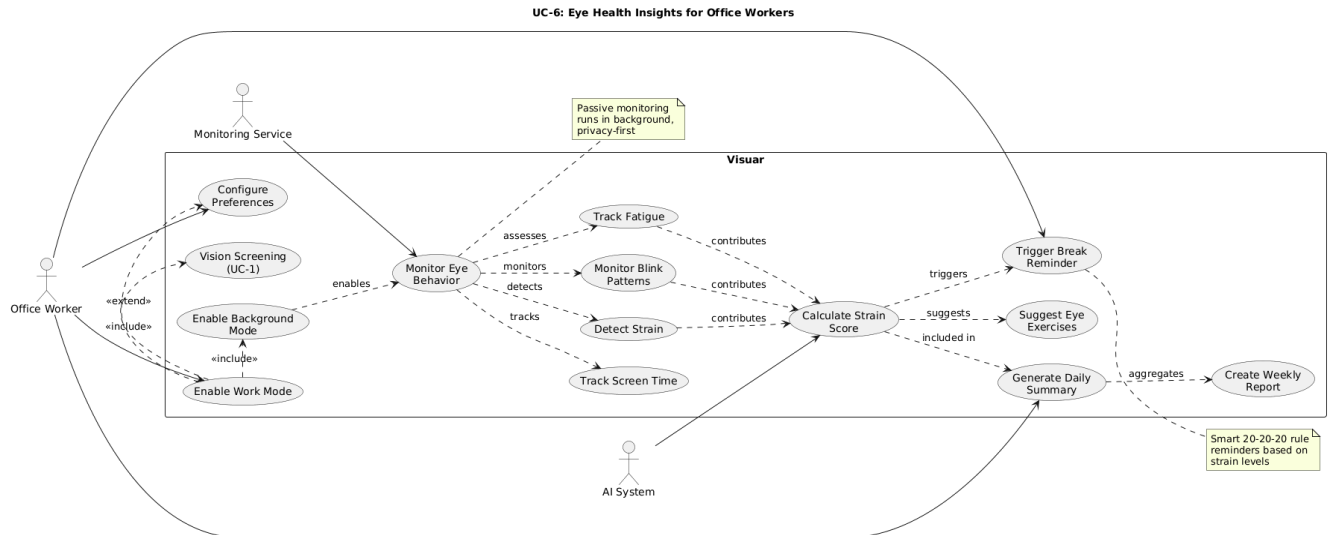


Figure 6: Eye Health Insights for Office Workers

Identifier	UC-6	
Purpose	Detect fatigue caused by prolonged screen exposure	
Priority	Medium	
Pre-conditions	System running in background (optional)	
Post-conditions	Fatigue insights generated	
Typical Course of Action		
S#	Actor Action	System Response
1	User enables "Work Mode Monitoring"	System tracks blink rate over time
2	User continues working	System detects reduced blink rate and dryness indicators
3	User receives alert	System recommends break or eye exercises
Alternate Course of Action		
S#	Actor Action	System Response

1	Webcam blocked	System disables monitoring and notifies user
2	User ignores alerts	System logs fatigue patterns for future use

Table 8: Eye Health Insights for Office Workers

4.7 Eye Health Report

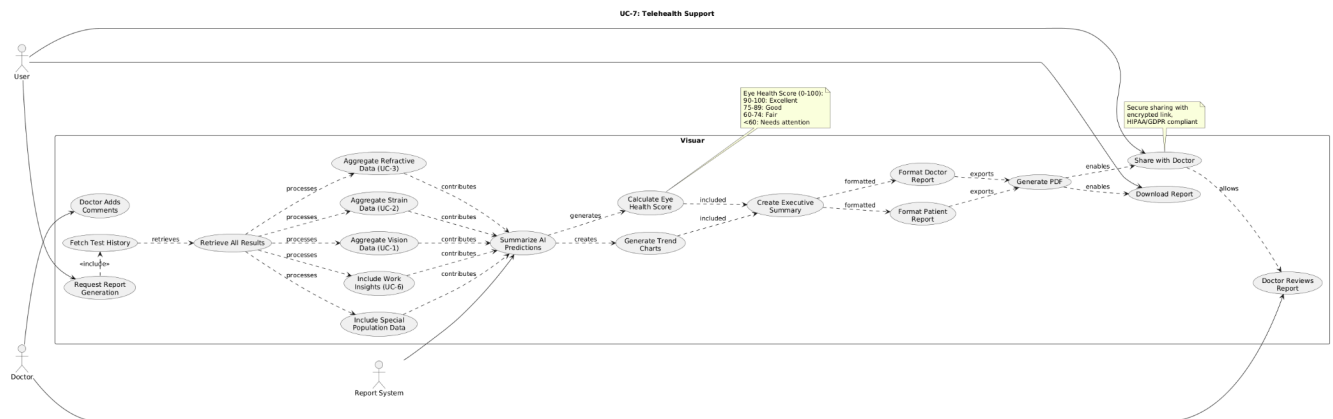


Figure 7: Telehealth Support

Identifier	UC-7	
Purpose	Generate doctor-friendly medical reports	
Priority	Medium	
Pre-conditions	User has completed at least one vision test	
Post-conditions	Report exported/downloaded	
Typical Course of Action		
S#	Actor Action	System Response
1	User selects "Generate Eye Health Report"	System compiles recent test results
2	User chooses download or share option	System exports formatted PDF/summary
3	User shares report with doctor	System confirms successful export
Alternate Course of Action		
S#	Actor Action	System Response

1	Missing data prevents report creation	System shows "Incomplete Test Data"
2	User cancels export	System returns to main dashboard

Table 9: Eye Health Report

4.8 Proposed Workflow

The proposed workflow of Visuar outlines the complete operational sequence of the system from user interaction to AI-assisted result generation. The workflow integrates all seven major use cases, demonstrating how the user, the webcam device, and the AI inference engine interact to perform the full vision-screening cycle.

1. User Initiation & Authentication

The user launches the system, logs in, and selects the desired vision assessment mode (general screening, child mode, elderly mode, refractive error test, or eye strain check).

2. Webcam Calibration & Environment Check

Before the assessment begins, the system performs webcam alignment, lighting validation, and face detection to ensure reliable data capture.

3. Live Video Capture

Once calibration is complete, the system captures real-time eye footage through the webcam. This step feeds into all subsequent analyses.

4. Feature Extraction

Computer vision modules extract eye landmarks, blink metrics, gaze direction, focus responses, and color-perception actions based on the selected test mode.

5. AI Evaluation

Extracted features are processed by trained machine-learning models to identify vision abnormalities such as refractive errors, eye strain indicators, attention deficits, and pattern-based risk markers.

6. Report Generation

The AI outputs are aggregated and interpreted to generate a structured eye-health report. This report includes risk flags, confidence scores, test summaries, and personalized recommendations.

7. Result Delivery

The final results are displayed to the user through a clean dashboard. Reports may be saved in the user profile or shared with a healthcare professional.

Visuar - Workflow Flowchart

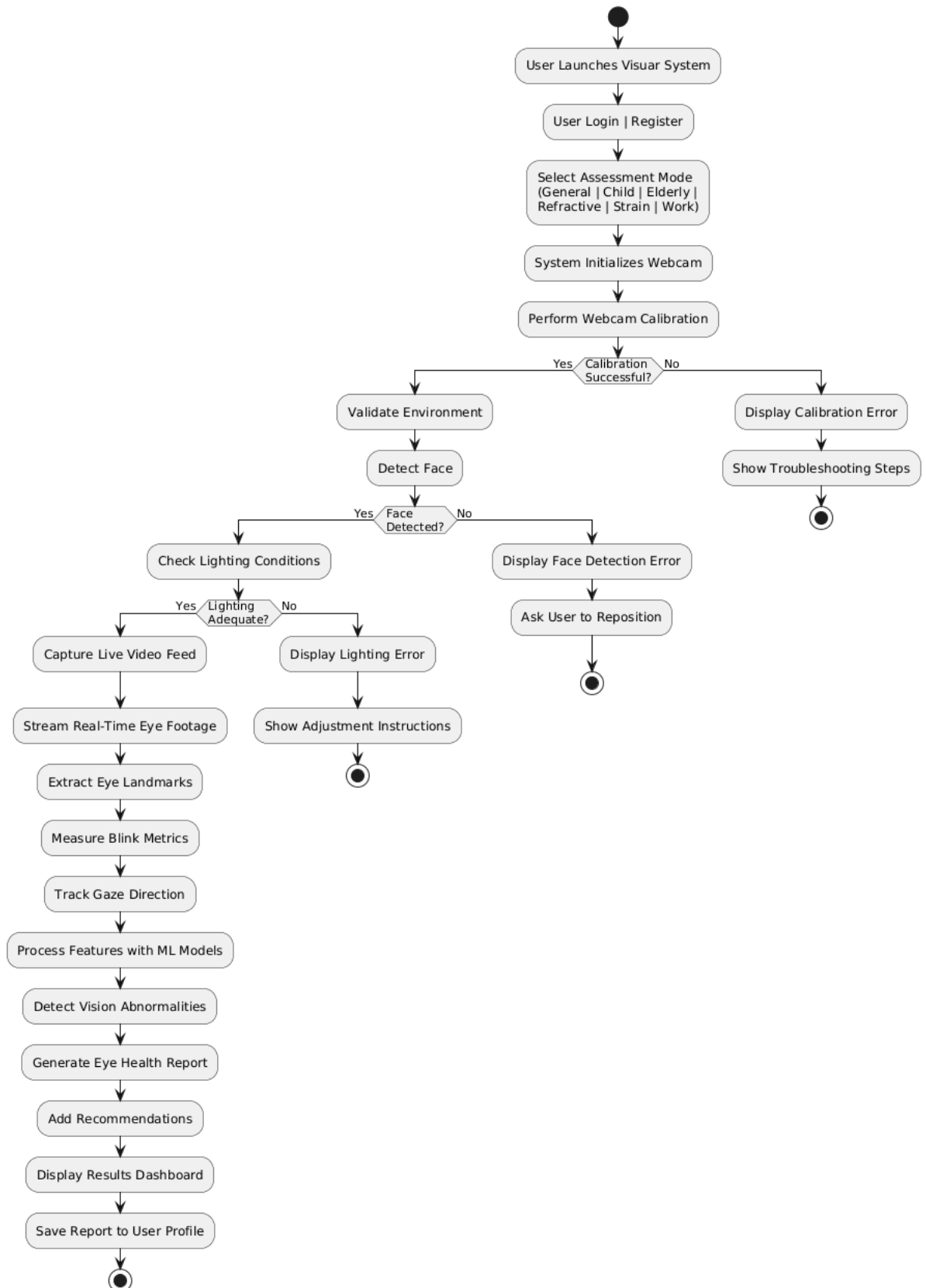


Figure 8: Visuar - Workflow Flowchart

This workflow ensures modularity, scalability, and consistency between all seven use cases, allowing Visuar to support multiple demographics and testing contexts using the same underlying pipeline.

4.9 Analysis and Modeling of Requirements

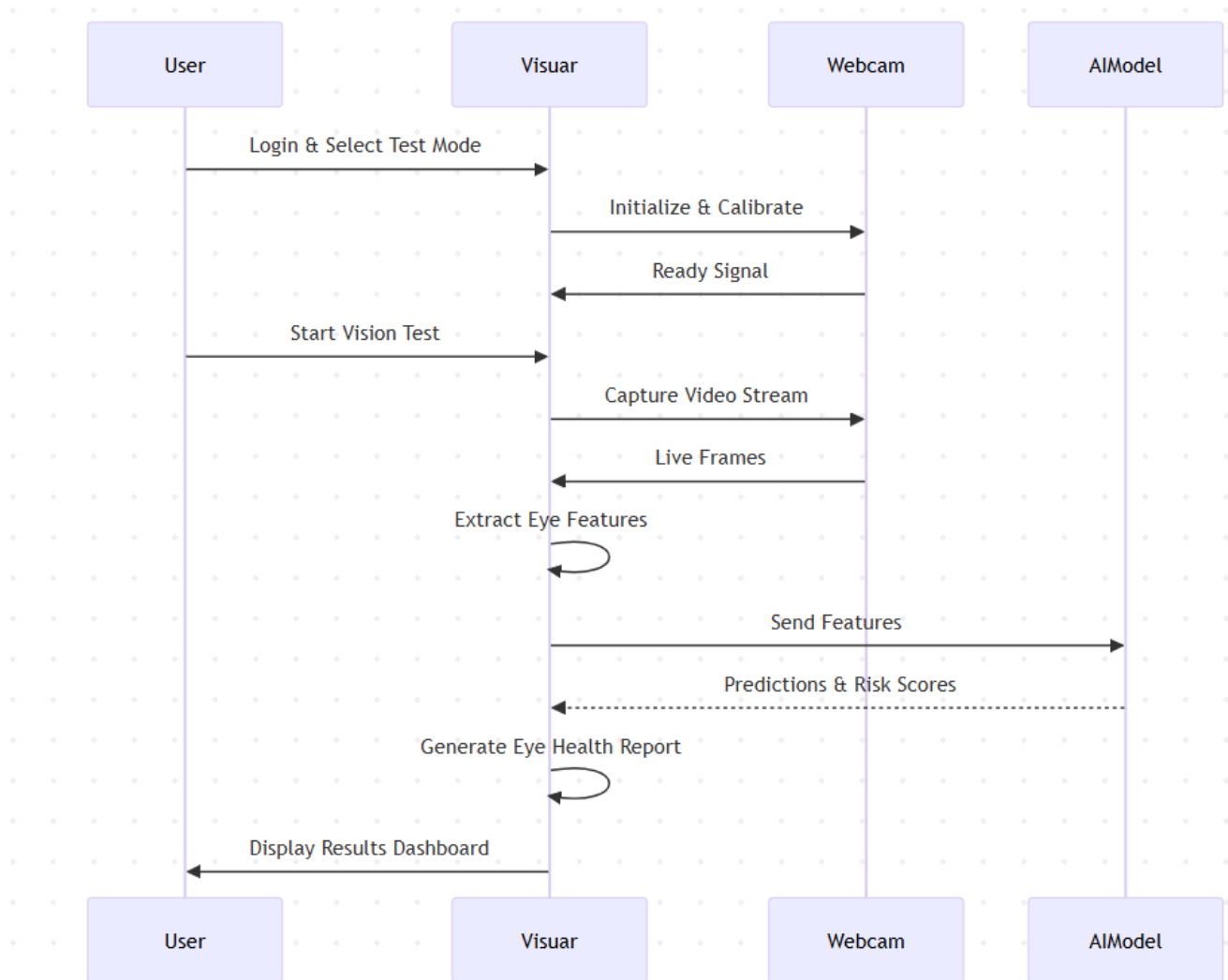


Figure 9: Sequence Diagram

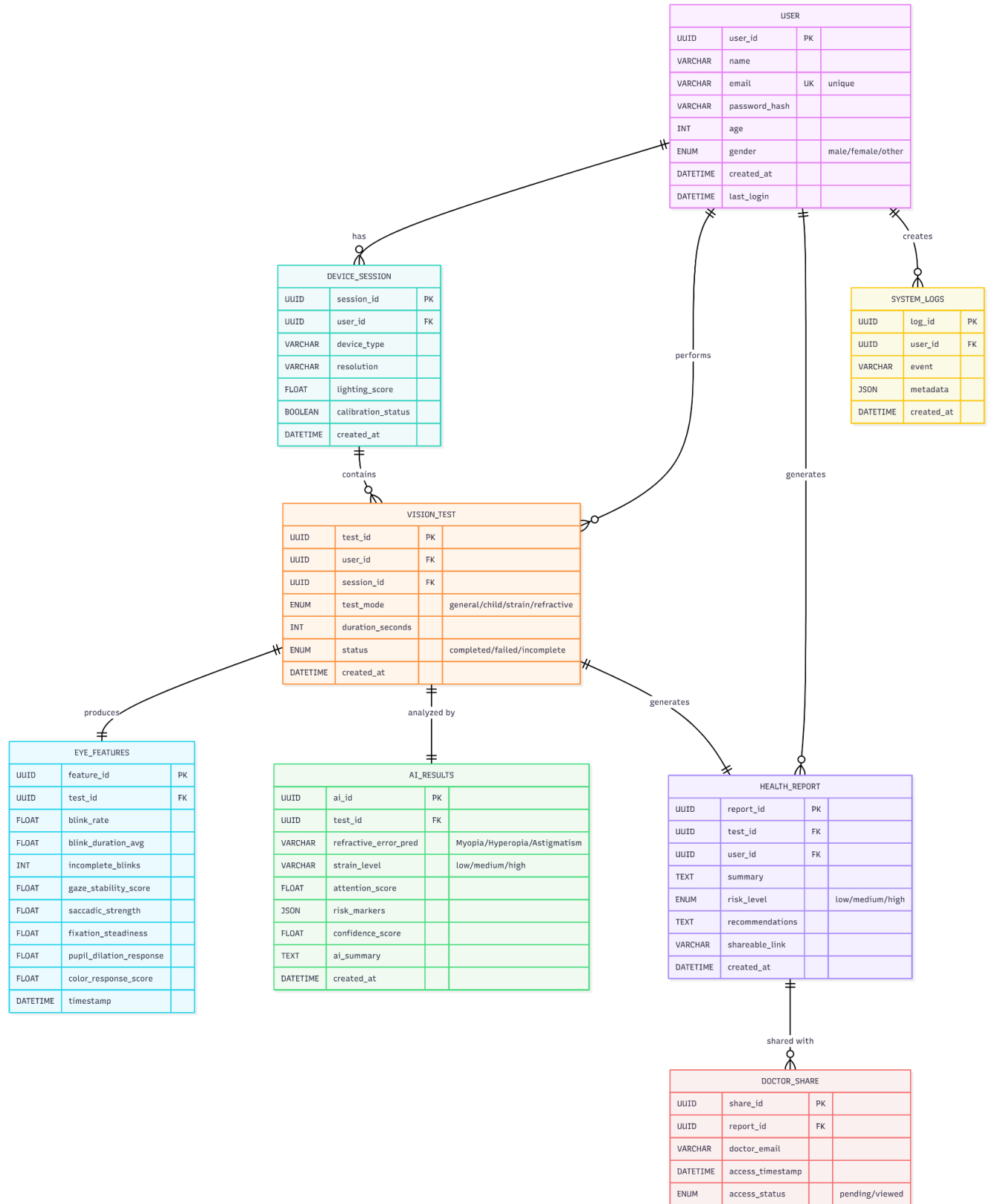


Figure 10: ERD Diagram — Visuar

DFD Level 0 - Visuar Context Diagram

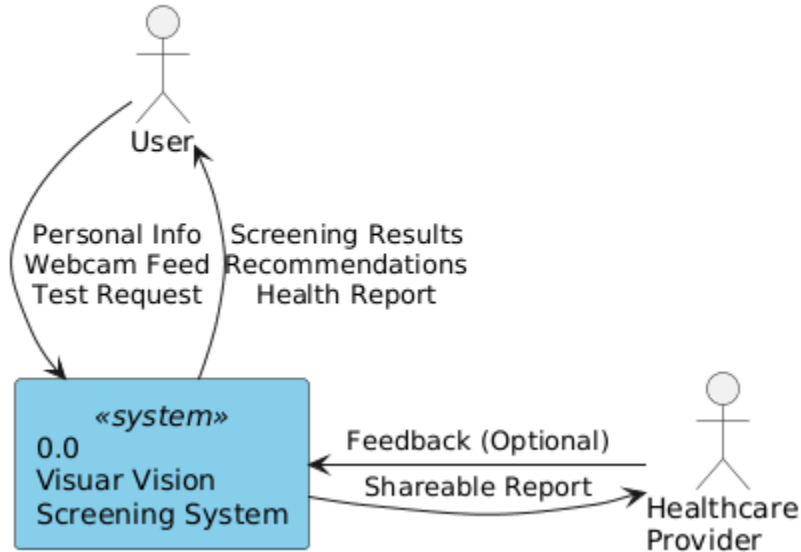


Figure 11: DFD 0 — Visuar

DFD Level 1 - Visuar Main Processes

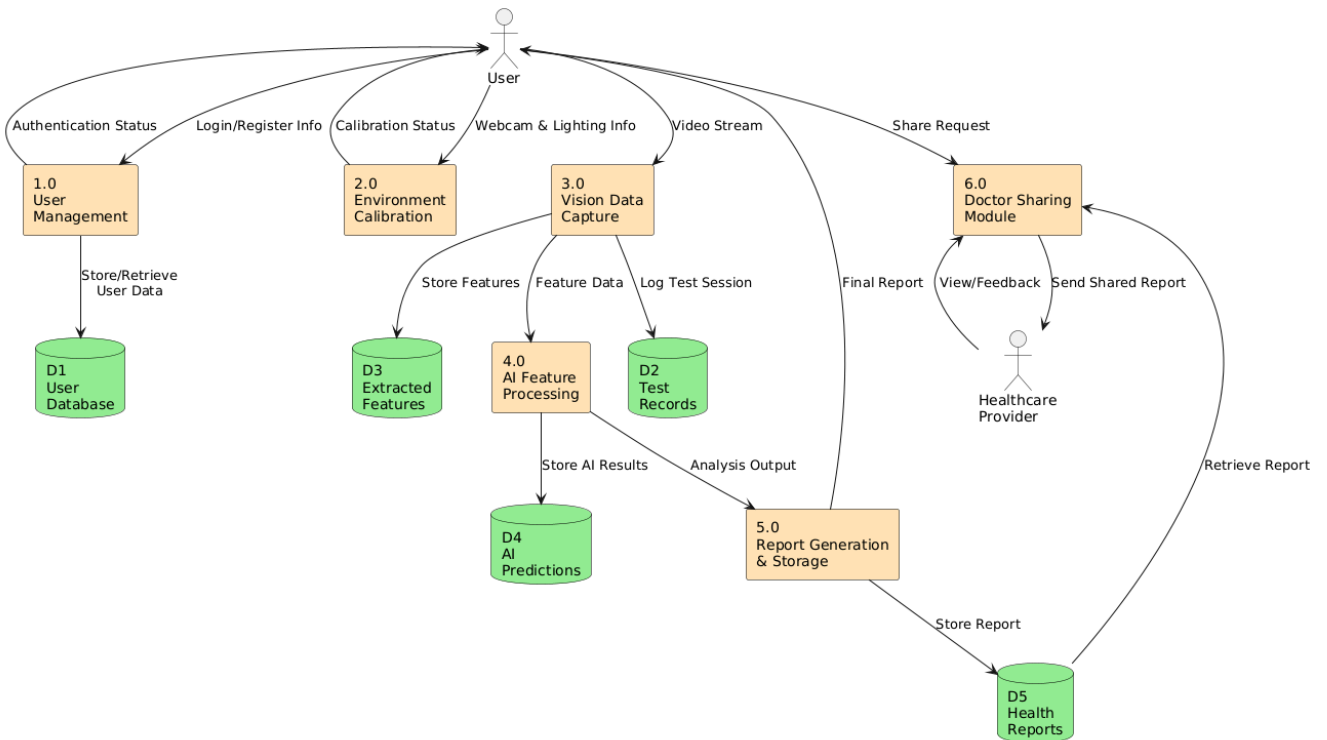


Figure 12: DFD 1 — Visuar

DFD Level 2 - Vision Data Capture (Process 3.0 Decomposed)

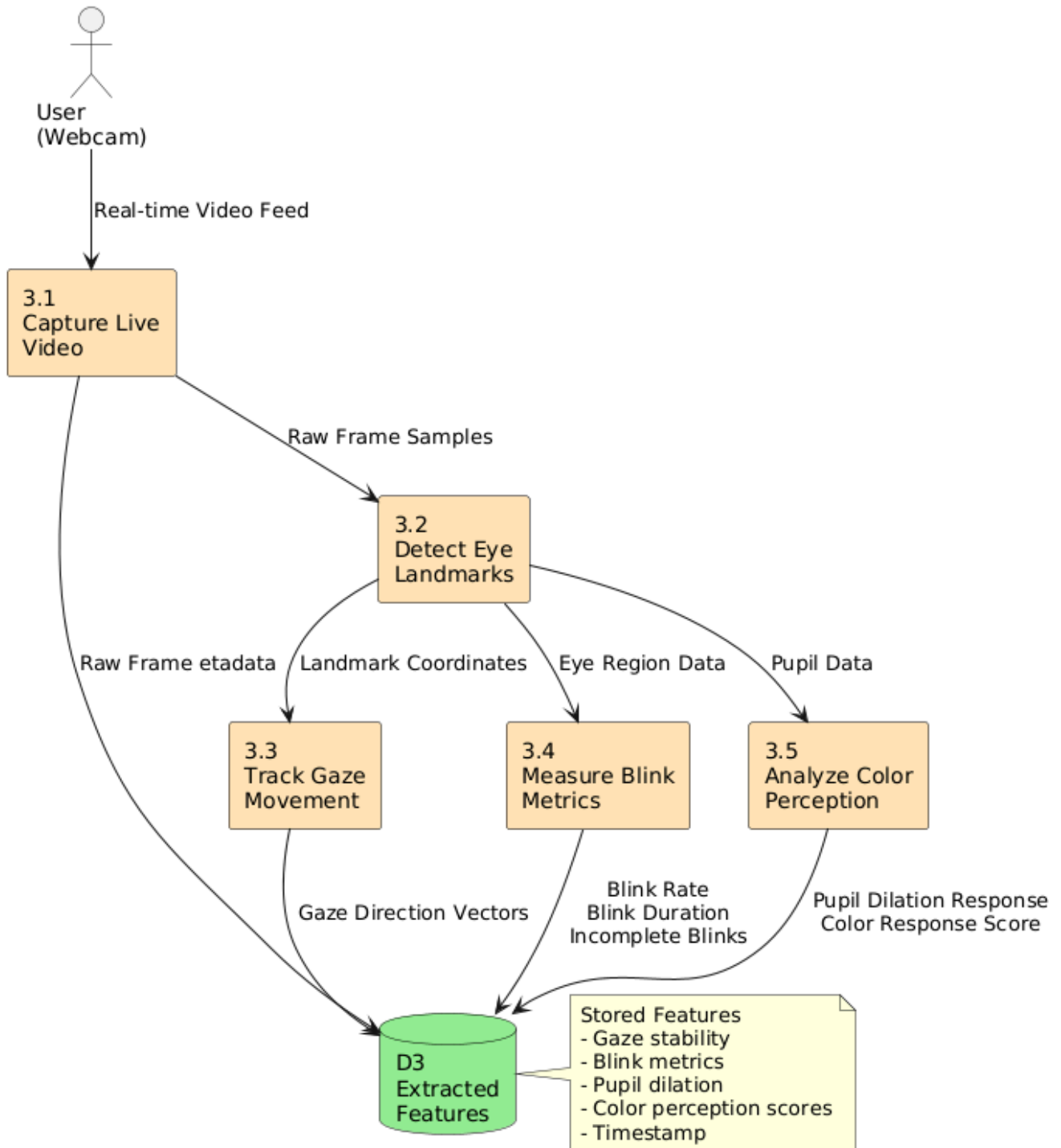
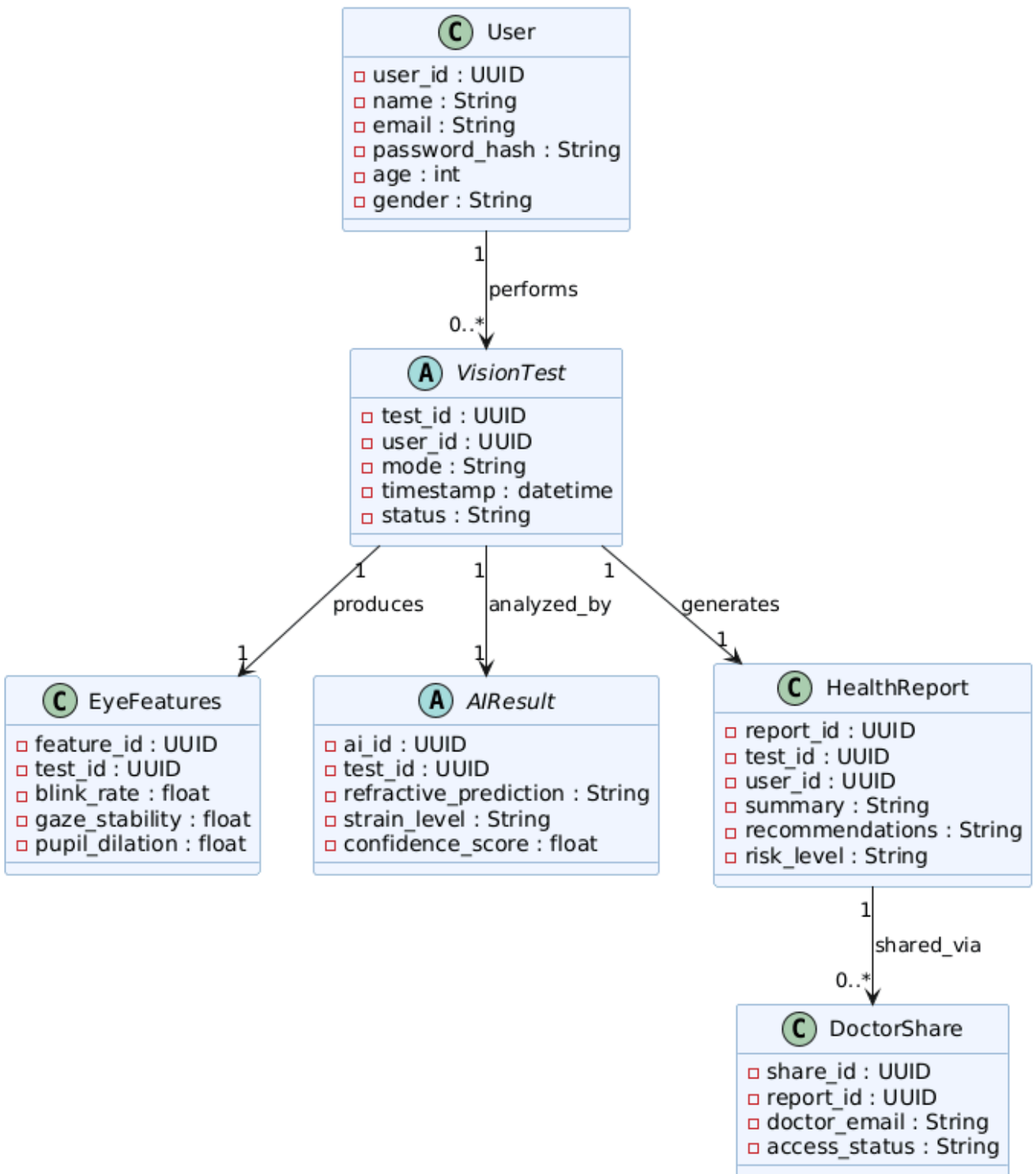


Figure 13: DFD 2 — Visuar

Visuar - Abstract Class Diagram**Figure 14: Abstract Class Diagram — Visuar**

5. Nonfunctional Requirements

5.1 Target Performance

The Visuar system should meet the following performance goals:

- **Real-time Processing:**
*Webcam video frames (10–30 FPS) must be processed with less than **150 ms latency** for smooth gaze, blink, and focus detection.*
- **Fast Model Inference:**
*Eye-landmark extraction, blink classification, and gaze estimation should complete within **200–300 ms** per frame on a standard laptop CPU (Intel i3/i5 or equivalent).*
- **Low Resource Consumption:**
The application must remain functional on devices with:
 - 4GB RAM
 - Integrated graphics
 - Built-in webcam (720p minimum)
- **Stable Output:**
*Health indicators (blink rate, gaze patterns, color-response scores) should achieve **≥85% consistency** across repeated tests under similar lighting.*

5.2 Safety Requirements

Although Visuar is a non-diagnostic wellness tool, the following safety considerations apply:

- **Eye Health Safety:**
The system must avoid exposing users to harmful light, strobing, or excessive brightness. All on-screen tests must comply with safe luminance guidelines.
- **User Guidance Safety:**
*The system must clearly state that results are for **preliminary guidance only** and do **not replace clinical evaluation**.*
- **Data Safety:**
No live camera feed is stored without explicit permission from the user.

5.3 Security Requirements

Visuar handles sensitive behavioral and biometric data; therefore, the system must ensure:

- **Encrypted Data Transfer:**
*All communication between modules (if cloud-based services are used) must use **HTTPS / TLS encryption**.*

- **Local-Only Processing (Default):**
All AI evaluations should run locally unless the user explicitly opts into cloud processing.
- **Secure Authentication:**
User login must support secure password hashing, session tokens, and timeout expiration.
- **Access Control:**
Only authenticated users may view their stored reports.
- **No Unauthorized Storage:**
Webcam frames must not be saved unless user explicitly checks “allow saving”.
- **Doctor-Share Privacy:**
Shared reports must be tokenized and expire automatically within 72 hours.

5.4 Additional Software Quality Attributes

Visuar must satisfy several quality attributes that ensure its reliability, trustworthiness, and usability as an AI-enabled eye-wellness system. The following characteristics are prioritized and defined in measurable, verifiable terms wherever possible.

Availability

- *The system should maintain **99% uptime** during normal usage on supported devices.*
- *All offline functionalities (webcam testing, local ML inference) must remain usable even if internet connectivity is unavailable.*

Correctness

- *Eye landmark detection accuracy should achieve **≥90% correctness** on internal validation tests.*
- *Interpretation logic (e.g., blink rate computation, color-detection scoring) must match clinical formulas or definitions with **≤5% deviation**.*

Flexibility

- *The user can toggle between modes (General, Child, Elderly, Strain, Refractive) without restarting the system.*
- *Text size and instruction pacing should auto-adjust based on selected mode.*

Interoperability

- *Visuar must be compatible with:*
 - *Built-in laptop webcams*
 - *USB webcams*
 - *Browser APIs (if web version is deployed)*

- Reports must be exportable in standard formats such as **PDF** and **JSON** for doctor review or integration in health platforms.

Maintainability

- Code should be organized into **independent modules** (Vision, ML, UI, Reporting).
- Modifying one module must not require changes to others (loose coupling).
- Developers should be able to implement minor enhancements within **≤2 working days**.

Portability

- The system must run on:
 - Windows 10+
 - macOS 12+
 - Web browser environments (Chrome, Firefox, Edge latest versions)
- Installation or setup time must not exceed **5 minutes** on a fresh device.

Reliability

- System accuracy (landmark extraction, blink detection) must remain functional across **95% of typical indoor lighting conditions**.
- Application should recover gracefully from webcam disconnection by auto-detecting and reinitializing the device.

Reusability

- Core modules (landmark extraction, model inference, preprocessing) must be reusable in at least **three different contexts**:
 - Wellness screening
 - Educational grading tools
 - Developer plugins for computer vision apps

Robustness

- The system must continue operating even when:
 - Lighting varies
 - User moves slightly
 - Backgrounds change
- In case of degraded input quality, the system must provide **clear corrective instructions** instead of failing silently.

Testability

- All modules must include clear input-output boundaries to allow unit testing.
- Automated test scripts should validate performance using **mock webcam frames** and sample datasets.
- Minimum test coverage must be **≥80%**.

Usability

- Users should be able to complete a full screening session in **under 3 minutes** without expert supervision.
- Instructions must be displayed in **simple, readable language**, with voice support optional.
- Elderly and child-friendly modes must use simplified screens, larger fonts, and minimal interaction steps.

Priority Preference Statement

For Visuar, the following priorities guide the quality attribute balance:

1. **Highest priority:** Usability, Correctness, Reliability
2. **Moderate priority:** Security, Adaptability, Maintainability
3. **Lower (but required) priority:** Portability, Reusability, Interoperability

These choices reflect Visuar's role as a **health-oriented wellness system that must be accurate, simple, and dependable** for non-technical users.

6. Other Requirements

- Lightning & Environment requirements
- Camera
- Medical Knowledge
- Data Privacy and Anonymization
- Data Integrity
- Real-time Feedback
- Multilingual Accessibility
- Microphone Accessibility
- Performance Logging & Analytics Extensibility for future Medical Modules

7. Initial Results

At this stage of development, the following initial results have been achieved:

7.1 Literature Review Completion

A detailed study of existing gaze estimation models, eye-tracking datasets, and clinical indicators of eye strain and refractive errors has been completed. Relevant research papers and datasets (RT-GENE, RT-BENE, NUIG EyeGaze01, Harvard FairVision subsets) were identified and documented.

7.2 Dataset Collection & Preparation

Multiple datasets have been successfully collected, downloaded, and organized, including:

- *Blink detection datasets [8]*
- *Gaze direction datasets [7, 11]*
- *Eye-region behavioral datasets [6]*
- *Disease-specific retinal datasets for extended experimentation [10]*

Preliminary preprocessing steps (frame extraction, cleaning, resizing) have been completed.

7.3 Prototype Vision Modules

Early experiments confirm that:

- *Webcam face detection and eye-region extraction work reliably in common lighting conditions*
- *Blink rate detection logic is functional on sample videos [8]*
- *Gaze direction models from RT-GENE/RT-BENE repositories load successfully [8, 11]*

7.4 Workflow & System Modeling

All major system diagrams — use cases, workflow chart, ERD, DFD, and class diagrams — have been drafted, representing the system architecture and data flow. These results validate the feasibility of Visuar and provide a stable foundation for the next development phase.

8. Revised Project Plan

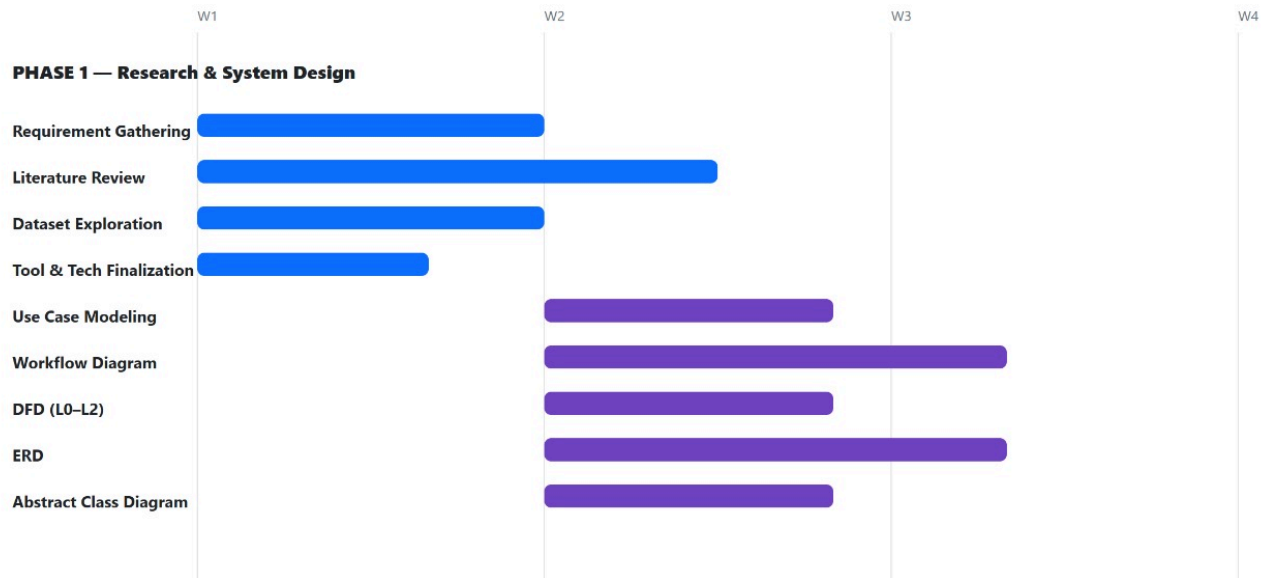


Figure 15: Gantt Chart

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Appendix A: Glossary

AI (Artificial Intelligence)

Technology enabling machines to perform tasks that typically require human intelligence.

API (Application Programming Interface)

A set of definitions that allow software components to communicate.

AMD (Age-Related Macular Degeneration)

An eye disease affecting the macula and causing vision loss; used for risk-pattern analysis.

Blink Rate

Number of blinks per minute; a biomarker for eye strain, attention, and fatigue.

Calibration

Initial system setup to align user position, lighting, and webcam orientation.

CNN (Convolutional Neural Network)

A deep learning model commonly used for image analysis and computer vision tasks.

Color Differentiation Test

A test that evaluates the user's ability to distinguish between similar colors.

Contrast Sensitivity

Assessment of how well a user detects low-contrast objects.

CV (Computer Vision)

A field of AI that extracts meaningful information from images and videos.

DFD (Data Flow Diagram)

A graphical representation of the flow of data within the system.

Diabetic Retinopathy (DR)

A diabetes-related eye disease characterized by retinal blood vessel damage.

Eye Landmark Detection

Identification of key points around the eyes (pupil, eyelid corners, iris center) for analysis.

Eye Tracking

Technology that determines where a user is looking in real time.

Fixation

A stable gaze on a single point; irregular patterns indicate eye strain or fatigue.

Focus Response

Speed and accuracy with which the eye shifts between near and far objects.

FPS (Frames Per Second)

Rate at which webcam frames are processed; important for real-time analysis.

Gaze Estimation

Predicting the direction of the user's gaze using ML models.

Gaze Stability

How consistently a user maintains gaze on a target; unstable patterns indicate strain or refractive issues.

Glaucoma

Optic nerve disease causing peripheral vision loss; early behavioral indicators may be detectable.

GUI (Graphical User Interface)

Visual components of the system enabling user interaction.

HCI (Human-Computer Interaction)

The field of designing usable and intuitive systems for human use.

Hyperopia (Farsightedness)

Difficulty focusing on near objects.

Ishihara Test

A color-blindness test used in the color differentiation module.

Latency

Delay between input capture (camera) and output response (AI model).

LLM (Large Language Model)

AI system used for natural-language recommendation generation.

MediaPipe

Google's CV framework used for face mesh and eye landmark extraction.

ML (Machine Learning)

Algorithms that learn patterns from data to make predictions or decisions.

Myopia (Nearsightedness)

Difficulty seeing distant objects clearly.

NFR (Non-Functional Requirement)

System qualities such as usability, performance, and reliability.

ONNX (Open Neural Network Exchange)

A lightweight model format used for running ML efficiently.

PDF Report

A downloadable document summarizing user test results and recommendations.

Presbyopia

Age-related inability to focus on near objects.

Pupil Detection

Locating the center of the pupil for gaze estimation.

Refractive Error

Mismatch in how the eye focuses light; includes myopia, hyperopia, and astigmatism.

RT-BENE

A dataset for blink estimation tasks.

RT-GENE

A dataset for gaze estimation tasks.

Saccades

Rapid eye movements between fixation points; abnormal patterns indicate strain.

Screen Distance Estimation

Process of approximate measurement of user-to-screen distance using CV.

SRS (Software Requirements Specification)

Formal document outlining the system requirements of Visuar.

STT (Speech-to-Text)

Technology converting user speech into text for interaction.

TTS (Text-to-Speech)

Technology used to provide verbal instructions and feedback.

UI/UX (User Interface / User Experience)

Visual layout and interaction flow of the system.

Webcam Calibration

Initial step ensuring proper lighting, face detection, and alignment for accurate results.

Wellness Recommendation Engine

LLM-based logic that interprets test results and provides advice.

Workflow Engine

Module managing the flow of tasks from interaction to reporting.

Appendix B: IV & V Report (Independent verification & validation)

IV & V Resource

Nayab Irfan



Name

Signature

S#	Defect Description	Origin Stage	Status	Fix Time (Hours)	Fix Time (Minutes)
1	Webcam fails to detect face if user has backlight	Prototype Testing	Fixed	0	40
2	Blink detection model misclassifies slow blinks as closed eyes	ML Model Integration	Pending	1	20
3	Text in PDF report overlaps on smaller screens	UI/Report Module	Fixed	0	55
4	Workflow stops abruptly if webcam reconnects during test	Integration Testing	In Progress	2	10
5	Color perception test inaccurate under warm lighting	Vision Processing	Pending	1	45
6	Gaze stability metric spikes when user slightly turns head	Algorithm Design	In Progress	1	30
7	Timer desynchronizes when switching between modes	UI Development	Fixed	0	35
8	Lighting warning appears even when lighting is adequate	Environment Evaluation Module	In Progress	1	00
9	PDF report fails to export when image attachment is added	Reporting Module	Fixed	0	55

10	Error messages not localized to Urdu in some screens	Localization System	Pending	0	50
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Table 10 : List of non-trivial defects

Appendix C – UML Diagram Explanations

This appendix provides detailed descriptions of all UML diagrams included in the Visuar system design. Each diagram is explained in terms of its purpose, major components, and how it contributes to the overall system architecture.

C.1 Use Case Diagram Explanation

The Use Case Diagram illustrates the interaction between the User and the Visuar system. Key use cases include:

- **Start Vision Test:** User initiates the webcam-based testing module.
- **Perform Eye Tracking:** The system detects gaze, pupil, and blink information.
- **Receive Test Feedback:** User receives real-time visual and verbal feedback.
- **Generate Eye Wellness Report:** System compiles results into a structured report.
- **Save or Share Report:** Users may export or save their diagnostic summary.

This diagram ensures a clear understanding of system boundaries and user-level behavior.

C.2 Activity Diagram Explanation

The Activity Diagram describes the sequential flow of processes during a vision test session:

1. **Initialization & Calibration**
 - Webcam activation, lighting check, and face alignment verification.
2. **Test Execution**
 - Gaze direction test
 - Blink/fatigue test
 - Focus stability test
 - Contrast/color test (where applicable)
3. **Data Processing**

- *ML inference (gaze + blink + stability metrics)*
- *Result scoring & classification*

4. **Report Generation & Completion**

- *AI-based recommendation synthesis*
- *Displaying results to the user*

This diagram highlights system concurrency (e.g., real-time ML inference running parallel to UI updates).

C.3 Sequence Diagram Explanation

The Sequence Diagram represents real-time interactions among system components:

- **User Interface → Camera Module:** Captures live frames.
- **Camera → Preprocessing Unit:** Crops eyes, normalizes frames.
- **Preprocessing → ML Engine:** Runs gaze/blink detection models.
- **ML Engine → Inference Layer:** Computes scores and interpretations.
- **Inference → LLM Recommendation Engine:** Generates personalized insights.
- **Engine → UI:** Displays results.

*This shows **timing**, **flow**, and **integration** between AI modules and UI components.*

C.4 Data Flow Diagram Explanation

*Shows the high-level flow: User → Input → Analysis Engine → Output.
Breaks the analysis engine into submodules:*

- *Pupil/Gaze Extraction*
- *Blink Detection*
- *Stability Measurement*
- *ML Classification*
- *Report Creation*

This confirms correct mapping between functional requirements and processing pipelines.

Appendix D – Dataset Metadata Sheet

This appendix summarizes all datasets used in the training, calibration, and validation of the Visuar system.

D.1 Primary Datasets

1. Eye Tracker Data (Shravan Bhat et al., 2023) [7]

- **Type:** Labeled eye images
- **Classes:** Left, Right, Up, Down, Center
- **Size:** 7,500 images
- **Format:** JPG images + structured folders
- **Purpose:** Train CNN gaze direction classifier
- **Source:** Mendeley Data

2. NUIG_EyeGaze01 Dataset (Kar et al., 2019) [6]

- **Type:** Numeric gaze angle dataset
- **Features:** Yaw, Pitch, Roll, Focus Deviation
- **Size:** 4 CSV files, multiple participants
- **Purpose:** Calibration + regression-based correction layer
- **Source:** Mendeley Data

3. RT-GENE Dataset (Imperial College London) [8,11]

- **Type:** Real-time gaze vectors & facial landmarks
- **Size:** Multiple sequences with annotations
- **Purpose:** High-precision gaze modeling
- **Source:** Zenodo

4. RT-BENE Dataset [8]

- **Type:** Blink-state labels (open/closed eyes)
- **Purpose:** Fatigue estimation and blink analysis
- **Source:** Zenodo

5. Harvard FairVision Dataset (AMD / DR / Glaucoma) [10]

- **Type:** Retinal fundus images (structural pathology)
- **Subsets:**
 - AMD folder → Age Related Macular Degeneration
 - DR folder → Diabetic Retinopathy
 - Glaucoma folder → Optic nerve damage
- **Purpose:** Medical reasoning layer, structural–behavioral correlation
- **Source:** Harvard Medical School (shared privately)

Appendix E – AI Model Specifications Table

This appendix summarizes the technical configuration and training details for all AI models inside Visuar.

E.1 Gaze Direction Classification Model

Attribute	Specification
Model Type	MobileNetV2 (Transfer Learning)
Input	Eye-region crop (224×224 RGB)
Output Classes	Left, Right, Up, Down, Center
Loss Function	Cross-Entropy
Optimizer	AdamW
Batch Size	32
Dataset Used	Eye Tracker Data (7,500 images) [7]
Accuracy Goal	85–95% test accuracy
Usage	Real-time gaze prediction

E.2 Blink Detection Model (RT-BENE Integration)

Attribute	Specification
Model Type	RT-BENE pretrained CNN [8]

<i>Input</i>	<i>Full eye frame (grayscale)</i>
<i>Output</i>	<i>Blink / No Blink</i>
<i>Purpose</i>	<i>Fatigue detection via blink rate (BPM)</i>
<i>Dataset</i>	<i>RT-BENE dataset [8]</i>

E.3 Gaze Vector Regression Model

Attribute	Specification
<i>Model Type</i>	<i>Random Forest Regression (NUIG_MLGaze) [12]</i>
<i>Input Features</i>	<i>Yaw, Pitch, Pupil Offset Metrics</i>
<i>Output</i>	<i>Continuous gaze angles</i>
<i>Dataset</i>	<i>NUIG_EyeGaze01 [6]</i>
<i>Purpose</i>	<i>Calibration for high accuracy</i>

E.4 Pupil Dilation Estimation Module

Attribute	Specification
<i>Model Type</i>	<i>OpenCV-based segmentation (heuristic)</i>
<i>Input</i>	<i>Eye frame</i>
<i>Output</i>	<i>Pupil diameter estimate</i>
<i>Purpose</i>	<i>Stress/fatigue and neurological cues</i>
<i>Dataset</i>	<i>PupilTracker sample data [14]</i>

E.5 Final Inference Layer (Health Scoring Engine)

Attribute	Specification
<i>Method</i>	<i>Hybrid rule-based + ML scoring</i>
<i>Inputs</i>	<i>Gaze stability, blink rate, pupil variation</i>

Visuar

<i>Outputs</i>	<i>Eye Wellness Score (0–100)</i>
<i>Purpose</i>	<i>Smart screening + recommendations via LLM</i>